

Numerical Analysis 7

1. Find the Lagrange polynomial that approximates $f(x) = x^3$.

(1) Find the linear interpolation polynomial $P_1(x)$ using the nodes $x_0 = -1$ and $x_1 = 1$.

Solution:

According to the question, we can get $f(x_0) = f(1) = 1, f(x_1) = f(-1) = -1$.

Construct the Lagrange bases:

$$L_{1,0}(x) = \frac{x - x_1}{x_0 - x_1} = \frac{x - 1}{-1 - 1} = -\frac{x - 1}{2}$$

$$L_{1,1}(x) = \frac{x - x_0}{x_1 - x_0} = \frac{x + 1}{1 - (-1)} = \frac{x + 1}{2}$$

$$P_1(x) = f(x_0)L_{1,0}(x) + f(x_1)L_{1,1}(x) = x$$

(2) Find the quadratic interpolation polynomial $P_2(x)$ using the the node $x_0 = -1, x_1 = 0$ and $x_2 = 1$.

Solution:

According to the question, we can get

$$f(x_0) = f(-1) = 1, f(x_1) = f(0) = f(-1), f(x_2) = f(1) = 1$$

$$L_{2,0}(x) = \frac{(x - 0)(x - 1)}{(-1 - 0)(-1 - 1)} = \frac{x^2 - x}{2}$$

$$L_{2,1}(x) = \frac{(x + 1)(x - 1)}{(0 + 1)(0 - 1)} = 1 - x^2$$

$$L_{2,2}(x) = \frac{(x + 1)(x - 0)}{(1 + 1)(1 - 0)} = \frac{x^2 + x}{2}$$

Thus,

$$P_2(x) = f(x_0)L_{2,0}(x) + f(x_1)L_{2,1}(x) + f(x_2)L_{2,2}(x) = x$$

(3) Find the cubic interpolation polynomial $P_3(x)$ using the nodes $x_0 = -1, x_1 = 0, x_2 = 1, x_3 = 2$.

Solution:

According to the question, we can get

$$f(x_0) = -1, f(x_1) = 0, f(x_2) = 1, f(x_3) = 8$$

$$L_{3,0}(x) = \frac{(x - 0)(x - 1)(x - 2)}{(-1 - 0)(-1 - 1)(-1 - 2)} = -\frac{x^3 - 3x^2 + 2x}{6}$$

$$L_{3,1}(x) = \frac{(x + 1)(x - 1)(x - 2)}{(0 + 1)(0 - 1)(0 - 2)} = \frac{x^3 - 2x^2 - x + 2}{2}$$

$$L_{3,2}(x) = \frac{(x + 1)(x - 0)(x - 2)}{(1 + 1)(1 - 0)(1 - 2)} = -\frac{x^3 - x^2 - 2x}{2}$$

$$L_{3,3}(x) = \frac{(x + 1)(x - 0)(x - 1)}{(2 + 1)(2 - 0)(2 - 1)} = \frac{x^3 - x}{6}$$

Thus,

$$P_3(x) = f(x_0)L_{3,0}(x) + f(x_1)L_{3,1}(x) + f(x_2)L_{3,2}(x) + f(x_3)L_{3,3}(x) = x^3$$

2. Let $f(x) = x + \frac{2}{x}$

(1) Use the quadratic interpolation polynomial based on the nodes $x_0 = 1, x_1 = 2$, and $x_2 = 2.5$ to approximate $f(1.5)$ and $f(1.2)$.

Solution:

we can easily get the $f(x_0) = 3, f(x_1) = 3, f(x_2) = 3.3$.

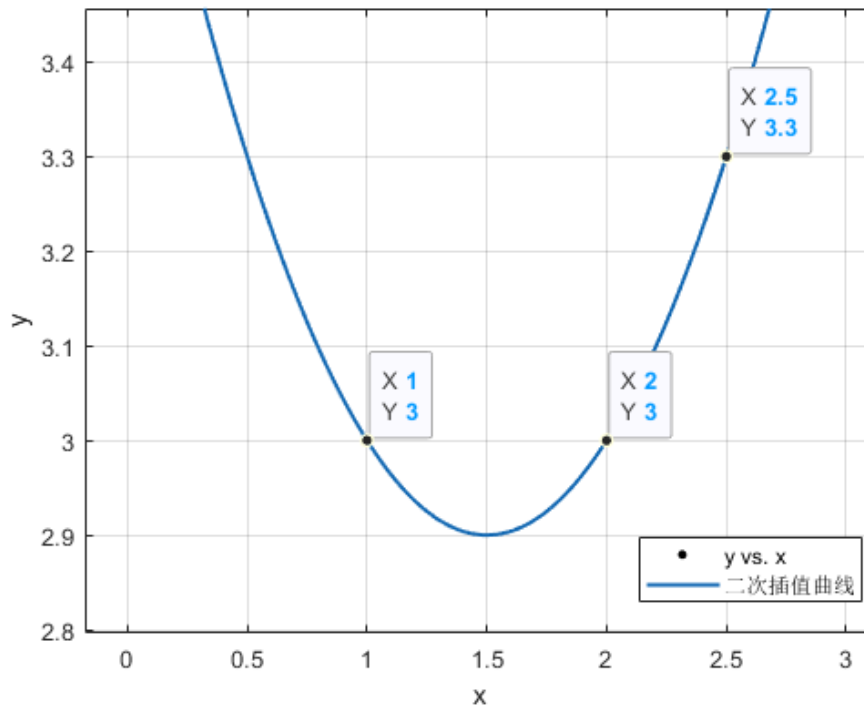
Construct the Lagrange bases:

$$\begin{aligned} L_{2,0}(x) &= \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)} = \frac{(x-2)(x-2.5)}{(1-2)(1-2.5)} = \frac{2(x-2)(x-2.5)}{3} \\ L_{2,1}(x) &= \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)} = \frac{(x-1)(x-2.5)}{(2-1)(2-2.5)} = -2(x-1)(x-2.5) \\ L_{2,2}(x) &= \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)} = \frac{(x-1)(x-2)}{(2.5-1)(2.5-2)} = 4(x-1)(x-2) \end{aligned}$$

the $P_2(x) = L_{2,0}(x)f(x_0) + L_{2,1}(x)f(x_1) + L_{2,2}(x)f(x_2) = 0.4x^2 - 1.2x + 3.8$

$f(1.5) = 2.9000, f(1.2) = 2.9360$.

The image of the interpolation function is as follows,



(2) Use the cubic interpolation polynomial based on the nodes $x_0 = 0.5, x_1 = 1, x_2 = 2$, and $x_3 = 2.5$ to approximate $f(1.5)$ and $f(1.2)$.

Solution:

we can easily get the $f(x_0) = 4.5, f(x_1) = 1, f(x_2) = 2, f(x_3) = 3.3$.

Construct the Lagrange bases:

$$\begin{aligned} L_{3,0}(x) &= \frac{(x-x_1)(x-x_2)(x-x_3)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)} = \frac{(x-1)(x-2)(x-2.5)}{(0.5-1)(0.5-2)(0.5-2.5)} \\ L_{3,1}(x) &= \frac{(x-x_0)(x-x_2)(x-x_3)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)} = \frac{(x-0.5)(x-2)(x-2.5)}{(1-0.5)(1-2)(1-2.5)} \\ L_{3,2}(x) &= \frac{(x-x_0)(x-x_1)(x-x_3)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)} = \frac{(x-0.5)(x-1)(x-2.5)}{(2-0.5)(2-1)(2-2.5)} \\ L_{3,3}(x) &= \frac{(x-x_0)(x-x_1)(x-x_2)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)} = \frac{(x-0.5)(x-1)(x-2)}{(2.5-0.5)(2.5-1)(2.5-2)} \end{aligned}$$

So the $f(x) = P_3(x) = L_{3,0}(x)f(x_0) + L_{3,1}(x)f(x_1) + L_{3,2}(x)f(x_2) + L_{3,3}(x)f(x_3) = -0.8x^3 + 4.8x^2 - 8.8x + 7.8$

$f(1.5) = 2.7000, f(1.2) = 2.7696.$

6. Let $f(x) = \sqrt{x}$.

(1) Compute the divided-difference table for f based on the nodes $x_0 = 4, x_1 = 5, x_2 = 6, x_3 = 7$, and $x_4 = 8$.

x_i	$f[]$	$f[,]$	$f[, ,]$	$f[, , ,]$	$f[, , , ,]$
4	2				
5	2.2361	0.2361			
6	2.4495	0.2134	-0.0113		
7	2.6458	0.1963	-0.0086	0.9143×10^{-3}	
8	2.8284	0.1827	-0.0068	0.5957×10^{-3}	-0.7965×10^{-4}

(2) Write the Newton's interpolation polynomial $P_1(x), P_2(x), P_3(x)$ and $P_4(x)$.

$$P_1(x) = f(x_0) + f[x_0, x_1](x - x_0) = 0.236(x - 4) + 2$$

$$P_2(x) = P_1(x) + f[x_0, x_1, x_2](x - x_0)(x - x_1) = 0.236(x - 4) + 2 - 0.0113(x - 4)(x - 5)$$

$$P_3(x) = P_2(x) + f[x_0, x_1, x_2, x_3](x - x_0)(x - x_1)(x - x_2) = 0.236x - 2.472 - 0.9143 \times 10^{-3}(x - 4)(x - 5)(x - 6)$$

$$P_4(x) = P_3(x) + f[x_0, x_1, x_2, x_3, x_4](x - x_0)(x - x_1)(x - x_2)(x - x_3) = 0.236(x - 4) + 2 - 0.7965 \times 10^{-4}(x - 4)(x - 5)(x - 6)(x - 7)$$

(3) Evaluate the polynomials in part(1) at $x = 4.5$ and $x = 7.5$.

Solution:

$$P_1(4.5) = 2.1181, P_1(7.5) = 2.8264$$

$$P_2(4.5) = 2.1209, P_2(7.5) = 2.7275$$

$$P_3(4.5) = 2.1209, P_3(7.5) = 2.7287$$

$$P_4(4.5) = 2.1204, P_4(7.5) = 2.7282$$

(4) Compare the values in part(3) with the actual function value of $f(x)$.

$$\text{actual value is } \sqrt{4.5} = 2.1213$$

$$\text{actual value is } \sqrt{7.5} = 2.7386$$